Indian Institute of Technology Patna

EC3202:Digital Signal Processing Lab

Experiment No. : 05

IIR Filter Design- Analog Filter

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Objective:

Aim 1:Design a lowpass Butterworth filter to satisfy

Passband cutoff: $\Omega p = 0.2\pi$;Passband ripple: $Rp = 7$ dB

Stopband cutoff: $\Omega s = 0.3\pi$;Stopband ripple: $As = 16$ dB

Aim 2:Design a lowpass Chebyshev-I filter to satisfy

Passband cutoff: $\Omega p = 0.2\pi$;Passband ripple: $Rp = 1$ dB

Stopband cutoff: $\Omega s = 0.3\pi$;Stopband ripple: $As = 16$ dB

Aim 3: Design a lowpass digital filter (Butterworth) using impulse invariance technique to satisfy

$\omega p = 0.2\pi, Rp = 1$ dB

$\omega s = 0.3\pi, As = 15$ dB

Aim 4:Design a lowpass digital filter (Chebyshev-I) using impulse invariance technique to satisfy

$\omega p = 0.2\pi, Rp = 1$ dB

$\omega s = 0.3\pi, As = 15$ dB

Aim 5: Design a digital filter (Butterworth) using Bilinear transformation technique to satisfy

$\omega p = 0.2\pi, Rp = 1$ dB

$\omega s = 0.3\pi, As = 15$ dB

Aim 6:Design a digital filter (Chebyshev-I) using Bilinear transformation technique to satisfy

$\omega p = 0.2\pi, Rp = 1$ dB

$\omega s = 0.3\pi, As = 15$ dB

Apparatus: MATLAB

Brief Theory:

Butterworth Lowpass Filters: This filter is characterized by the property that its magnitude is flat in both passband and stopband. The magnitude squared response of an Nth-order lowpass filter is given by:

$$|H_a(j\Omega)|^2 = \frac{1}{1 + (\frac{\Omega}{\Omega_c})^{2N}}$$

where N is the order of the filter and Ωc is the cutoff frequency in rad/sec.

Design Equations :The analog lowpass filter is specified by the parameters Ωp, Rp, Ωs and As Therefore the essence of the design in the case of Butterworth filter is to obtain the order N and the cutoff frequency Ωc, given these specifications. We want

- at $\Omega = \Omega_p$, $-10\log_{10}|H_a(j\Omega)|^2 = R_p$ or

$$-10\log_{10}(\frac{1}{1 + (\frac{\Omega_p}{\Omega_c})^{2N}}) = R_p \tag{2}$$

and

- at $\Omega = \Omega_s$, $-10\log_{10}|H_a(j\Omega)|^2 = A_s$ or

$$-10\log_{10}(\frac{1}{1 + (\frac{\Omega_s}{\Omega_c})^{2N}}) = A_s \tag{3}$$

Solving these two equations for N and Ω_c, we have

$$N = \frac{\log_{10}[(10^{R_p/10} - 1)/(10^{A_s/10} - 1)]}{2\log_{10}(\Omega_p/\Omega_s)} \tag{4}$$

In general, the above N will not be an integer. Since we want N to be an integer, we must choose

$$N = \frac{\log_{10}[(10^{R_p/10} - 1)/(10^{A_s/10} - 1)]}{2\log_{10}(\Omega_p/\Omega_s)} \tag{5}$$

where the operation [x] means "choose the smallest integer larger than x." Since the actual N chosen is larger than required, specifications can be either met or exceeded either at Ω_p or at Ω_s. To satisfy the specifications exactly at Ω_p,

$$\Omega_c = \frac{\Omega_p}{2N\sqrt{(10^{R_p/10} - 1)}} \tag{6}$$

or, to satisfy the specifications exactly at Ω_s,

$$\Omega_c = \frac{\Omega_s}{2N\sqrt{(10^{A_s/10}-1)}} \tag{7}$$

Chebyshev Lowpass Filters:

The magnitude-squared response of a chebyshev-I filter is

$$|H_a(j\Omega)|^2 = \frac{1}{1+\epsilon^2 T_N^2(\frac{\Omega}{\Omega_c})} \tag{8}$$

where N is the order of the filter, is the passband ripple factor, which is related to R_p, and $T_N(x)$ is the Nth-order

Chebyshev polynomial
given by

$$\mathrm{T}_N(x) = \begin{cases} \cos(N\cos^{-1}(x)), & 0 \le x \le 1 \\ \cosh(\cosh^{-1}(x)), & 1 < x < \infty, \text{where } x = \frac{\Omega}{\Omega_c} \end{cases}$$

The Equi ripple response of the Chebyshev filters is due to this polynomial $T_N(x)$. Its key properties are

(a) for $0 < x < 1$, $T_N(x)$ Oscillates between -1 and 1, and

(b) for $1 < x < \infty$, $T_N(X)$ increases monotonically to ∞.

Design Equations

Given Ω_p, Ω_s, R_p and A_s, three parameters are required to determine a Chebyshev-I filter: ϵ, Ω_c and N. we obtain

$$\epsilon = \sqrt{10^{0.1R_p}-1} \text{ and } A = 10As/20$$

from the properties discussed above we have

$$\Omega_c = \Omega_p \text{ and } \Omega_r = \frac{\Omega_s}{\Omega_p}$$

the order N is given by

$$g = \sqrt{(A^2-1)/\epsilon^2} \tag{9}$$

$$N = \frac{log_{10}[g + \sqrt{g^2 - 1}]}{log_{10}[\Omega_r + \sqrt{\Omega_r^2 - 1}]} \tag{10}$$

Design Procedure for impulse invariable Technique:

Given the digital lowpass filter specifications ω_p, ω_s, R_p and A_s, we want to determine $H(s)$ by first designing an equivalent analog filter and then mapping it into the desired digital filter. The steps required for this procedure are 1. Choose T and determine the analog frequencies

$$\Omega_p = \frac{\omega_p}{T_p} \text{ and } \Omega_s = \frac{\omega_s}{T}$$

2. Design an analog filter $H_a(s)$ using the specifications Ω_p, Ω_s, R_p and A_s. This can be done using anyone of the three (Butterworth, Chebyshev, or elliptic) prototypes of the previous section.

3. Using partial fraction expansion, expand $H_a(s)$ into

$$H_a(s) = \sum_{k=1}^{N} \frac{R_k}{s - p_k} \tag{11}$$

4. Now transform analog poles p_k into digital poles e^{pkT} to obtain the digital filter:

$$H_a(s) = \sum_{k=1}^{N} \frac{R_k}{1 - e^{p_k T} z^{-1}} \tag{12}$$

Design Procedure for Bi-linear transformation Technique:

Given the digital lowpass filter specifications ω_p, ω_s, R_p and A_s, we want to determine $H(z)$. The design steps in the procedure are the following:

1. Choose T. This is arbitrary, and we may set $T = 1$.

2. Prewrap the cutoff frequencies ω_p and ω_s; that is, calculate Ω_p and Ω_s,

$$\Omega_p = \frac{2}{T} tan\left(\frac{\omega_p}{2}\right),\ \Omega_s = \frac{2}{T} tan\left(\frac{\omega_s}{2}\right)$$

3. Design an analog filter $H_a(s)$ using the specifications Ω_p, Ω_s, R_p and A_s.

4. Finally, set

$$H(z) = H_s\left(\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}\right) \tag{13}$$

and simplify to obtain $H(z)$ as a rational function in z^{-1}.

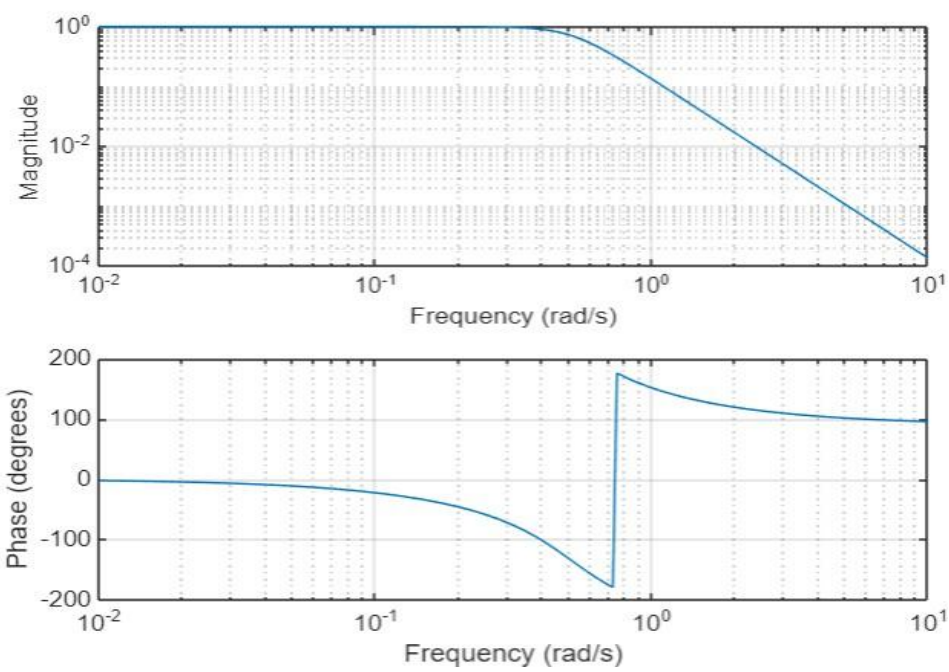
Results:

Aim 1:

```
% Aim 1
wp=0.2*pi;
Rp=7;
ws=0.3*pi;
As=16;
[n,Wc]=buttord(wp,ws,Rp,As,'s')
[b,a]=butter(n,Wc,'s');
freqs(b,a);
```

```
n = 3

Wc = 0.5122
```

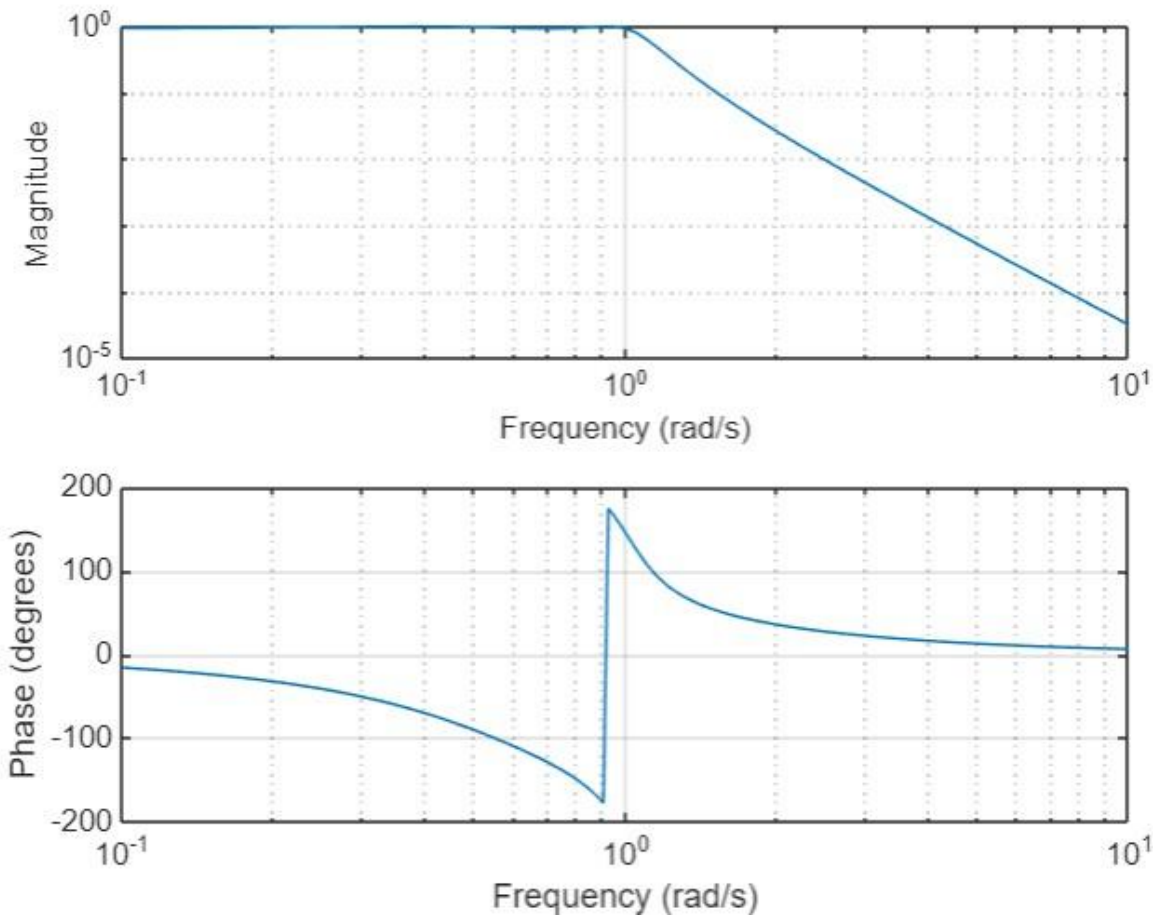


Aim 2:

```
% Aim2
wp=0.2*pi;
Rp=1;
ws=0.3*pi;
As=16;
[n,wp]=cheb1ord(wp,ws,Rp,As,'s')
[b,a]=cheby1(n,wp,Rp,'s');
freqs(b,a);
```

```
n = 4

wp = 0.6283
```

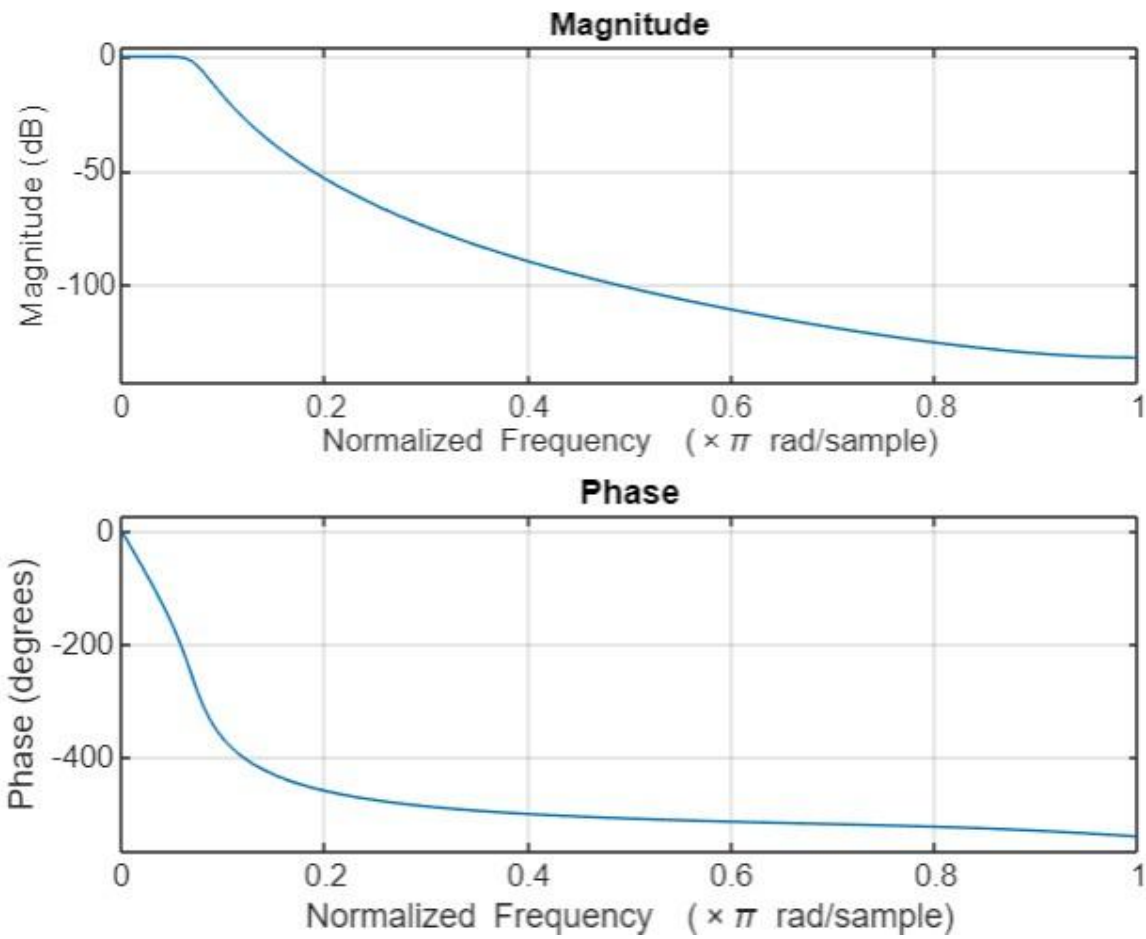


Aim 3:

```
% Aim 3
wp=0.2;
Rp=1;
ws=0.3;
As=15;
[n,Wc]=buttord(wp,ws,Rp,As,'s')
[b,a]=butter(n,Wc,'s');
[bz,az]=impinvar(b,a,1);
freqz(bz,az);
```

```
n = 6

Wc = 0.2256
```

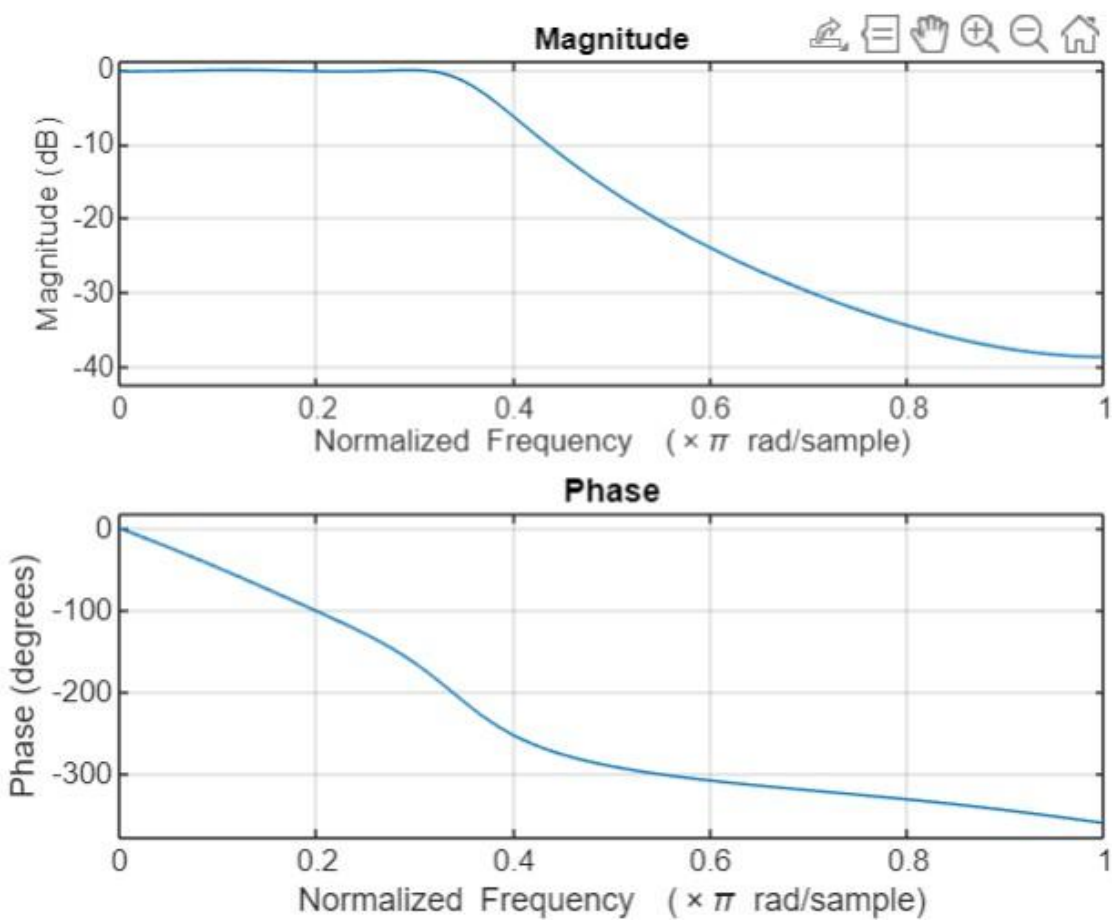


Aim 4:

```
% aim 4
wp=0.2;
Rp=1;
ws=0.3;
As=15;
[n,wp]=cheb1ord(wp,ws,Rp,As,'s')
[b,a]=cheby1(n,wp,Rp,'s')
[bz,az]=impinvar(b,a,1);
freqz(bz,az)
```

```
n = 4

wp = 0.2000
```

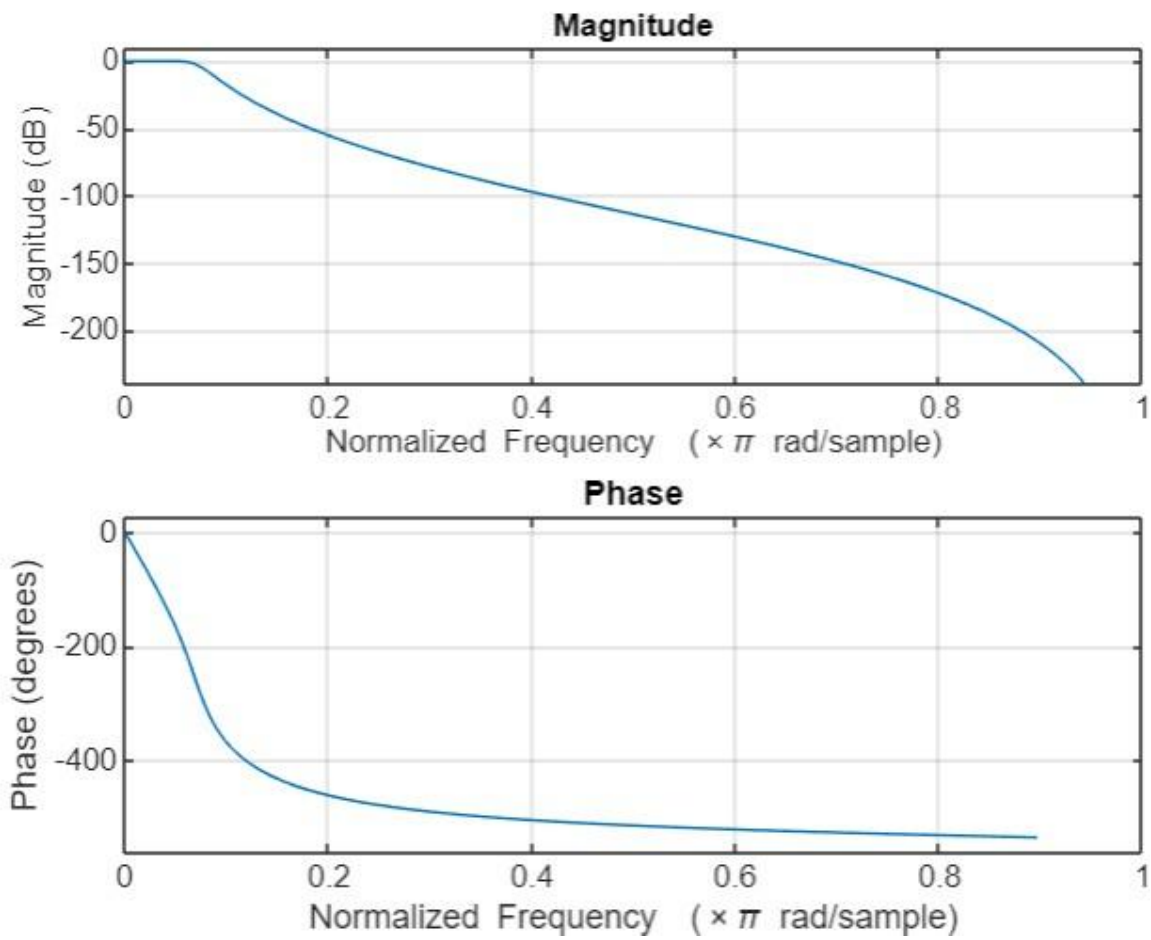


Aim 5:

```
% aim 5
wp=0.2;
Rp=1;
ws=0.3;
As=15;
[n,Wc]=buttord(wp,ws,Rp,As,'s')
[b,a]=butter(n,Wc,'s');
[bz,az]=bilinear(b,a,1);
freqz(bz,az)
```

```
n = 6

Wc = 0.2256
```

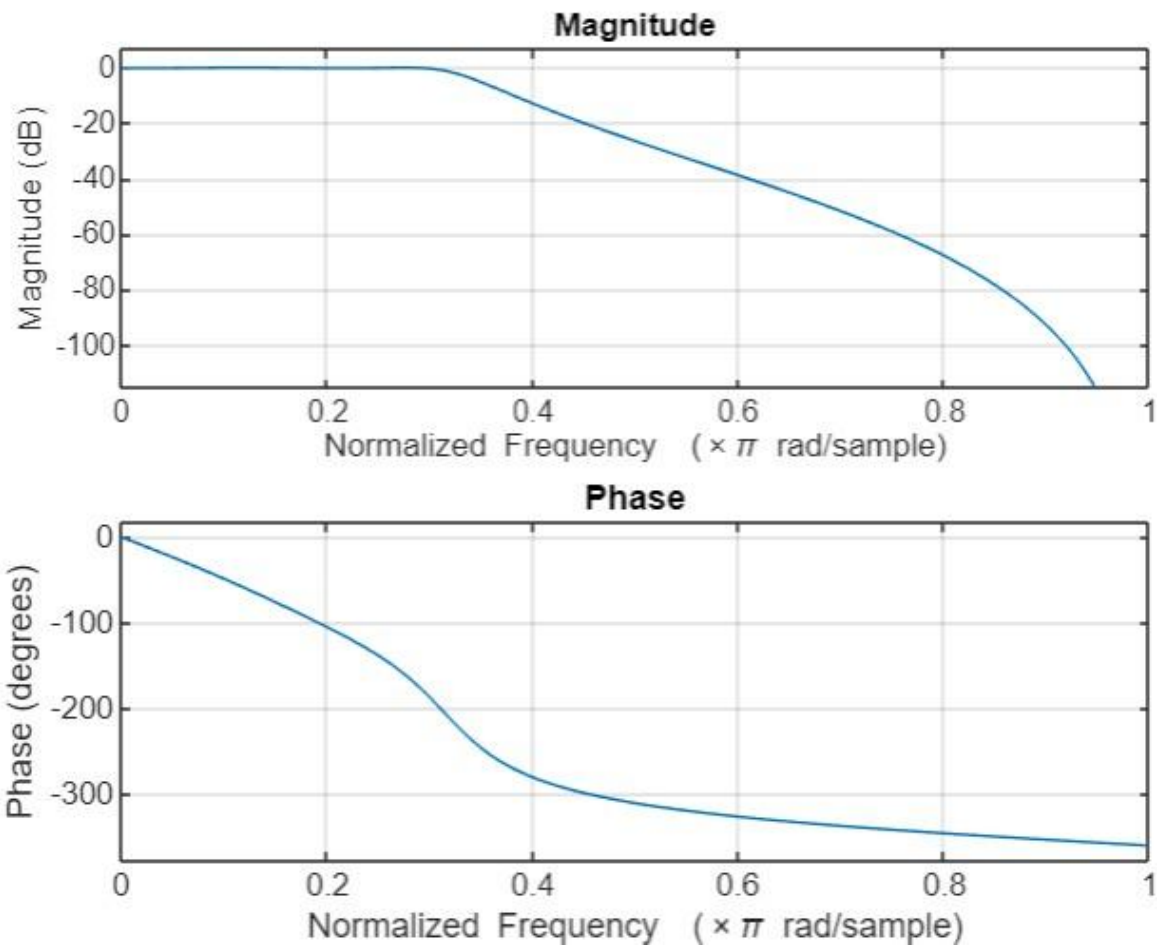


Aim 6:

```
% aim 6
wp=0.2;
Rp=1;
ws=0.3;
As=15;
[n,wp]=cheb1ord(wp,ws,Rp,As,'s')
[b,a]=cheby1(n,wp,Rp,'s');
[bz,az]=bilinear(b,a,1);
freqz(bz,az)
```

```
n = 4

wp = 0.2000
```



Conclusion:

In this experiment, analog lowpass filters (Butterworth and Chebyshev-I) were successfully designed using given specifications of passband and stopband frequencies along with ripple constraints. The Butterworth filter provided a smooth and flat frequency response, while the Chebyshev filter achieved a sharper transition at the cost of passband ripples.

Further, digital filters were designed using both impulse invariance and bilinear transformation techniques. It was observed that impulse invariance preserves the time-domain characteristics but may introduce aliasing, whereas bilinear transformation avoids aliasing by frequency warping and provides better control over the frequency response.

Overall, the experiment demonstrated the trade-offs between different filter types and design techniques, helping in understanding practical filter design and implementation in digital signal processing systems.